



Hybrid RAG Knowledge Assistant

A Privacy-First Intelligent Document Question Answering System

MAJOR PROJECT

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Why This System?

The LLM Gap

LLMs rely on static training data – no access to enterprise documents, risking hallucinations

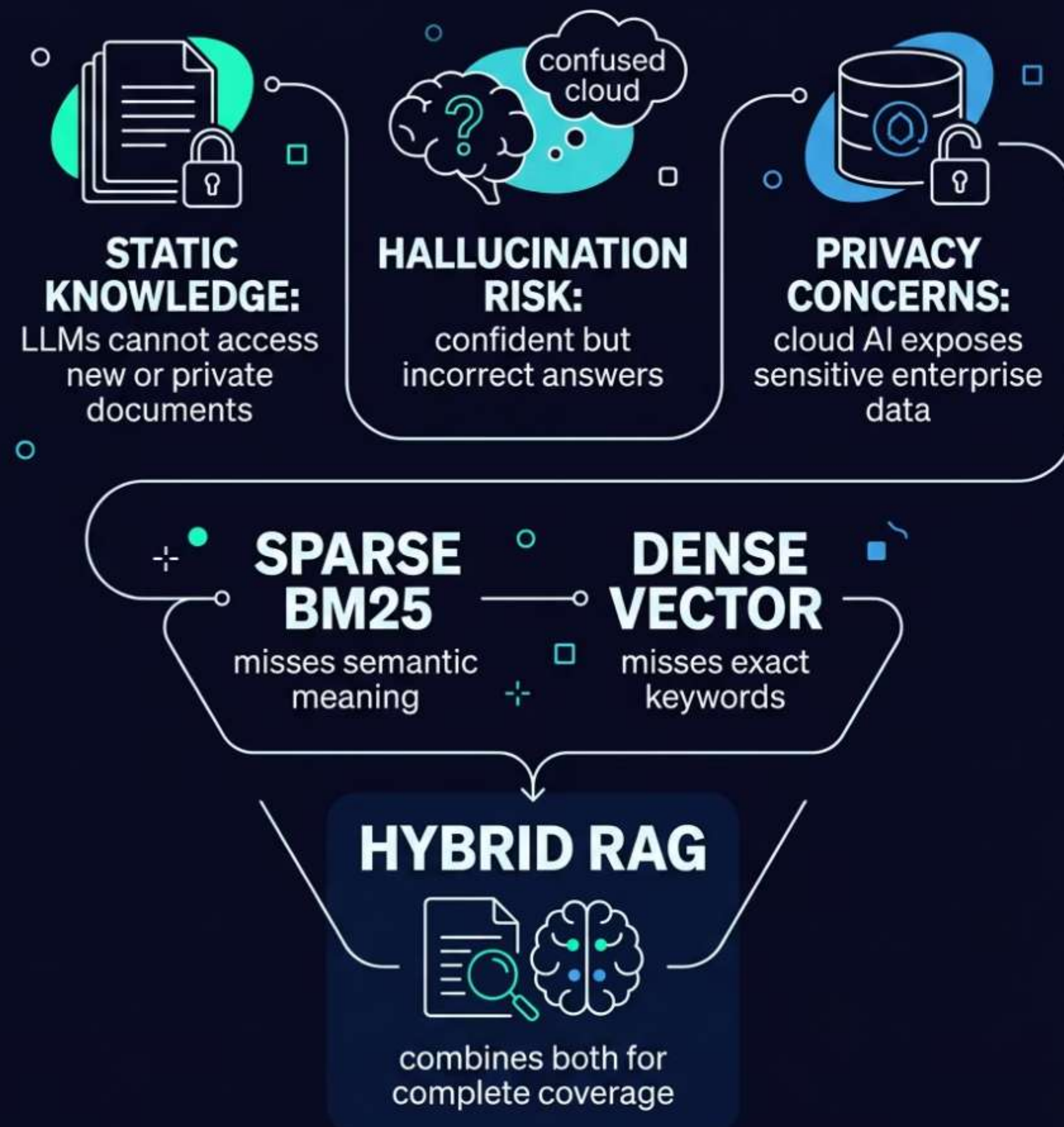
The Privacy Risk

Cloud-based AI exposes sensitive organizational data

The Need

Accurate, privacy-preserving, real-time document QA

Problem & Research Gap



Existing Challenges

- Static knowledge in LLMs
- Hallucinated responses
- Limited enterprise document access
- Privacy concerns with cloud AI

Research Gap

Existing RAG systems rely on either **BM25** (sparse) or **vector search** (dense) – not both.

Our Solution: Hybrid RAG combining semantic + keyword retrieval

Project Objectives

1

Reduce Hallucinations

Source-cited, document-grounded responses

2

Online & Offline AI

Ollama (local) + OpenAI / Groq / Anthropic (cloud)

3

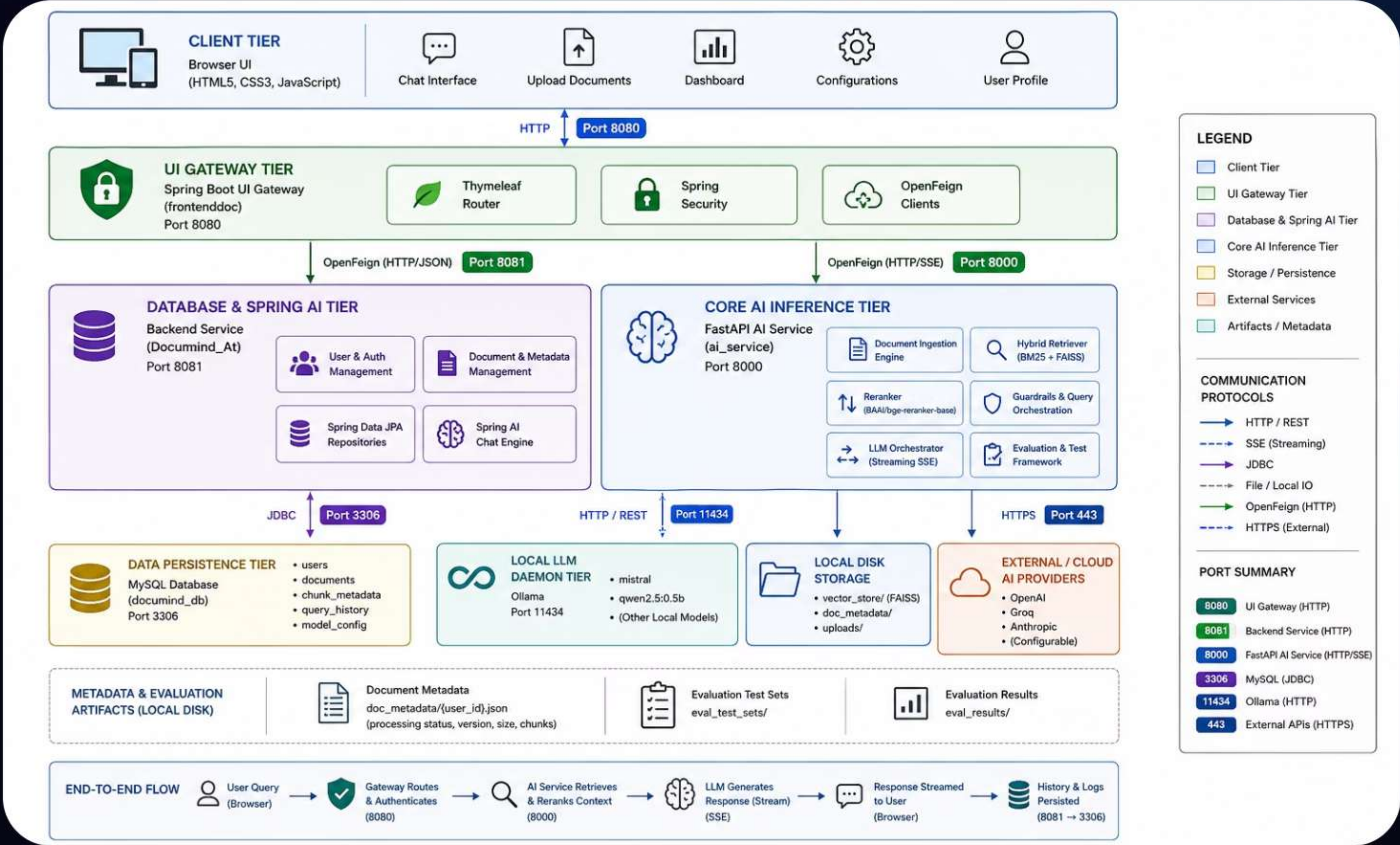
Document Privacy

Secure storage with access control

Technology Stack

Layer	Technologies
Frontend	HTML5, CSS3, JavaScript, Bootstrap
Backend	Java 21, Spring Boot, Spring Security
AI Service	Python, FastAPI, LangChain
Database	MySQL
Embeddings	all-MiniLM-L6-v2
Retrieval	BM25 + FAISS
LLM	Ollama (local) · OpenAI / Groq / Anthropic (cloud)

System Architecture

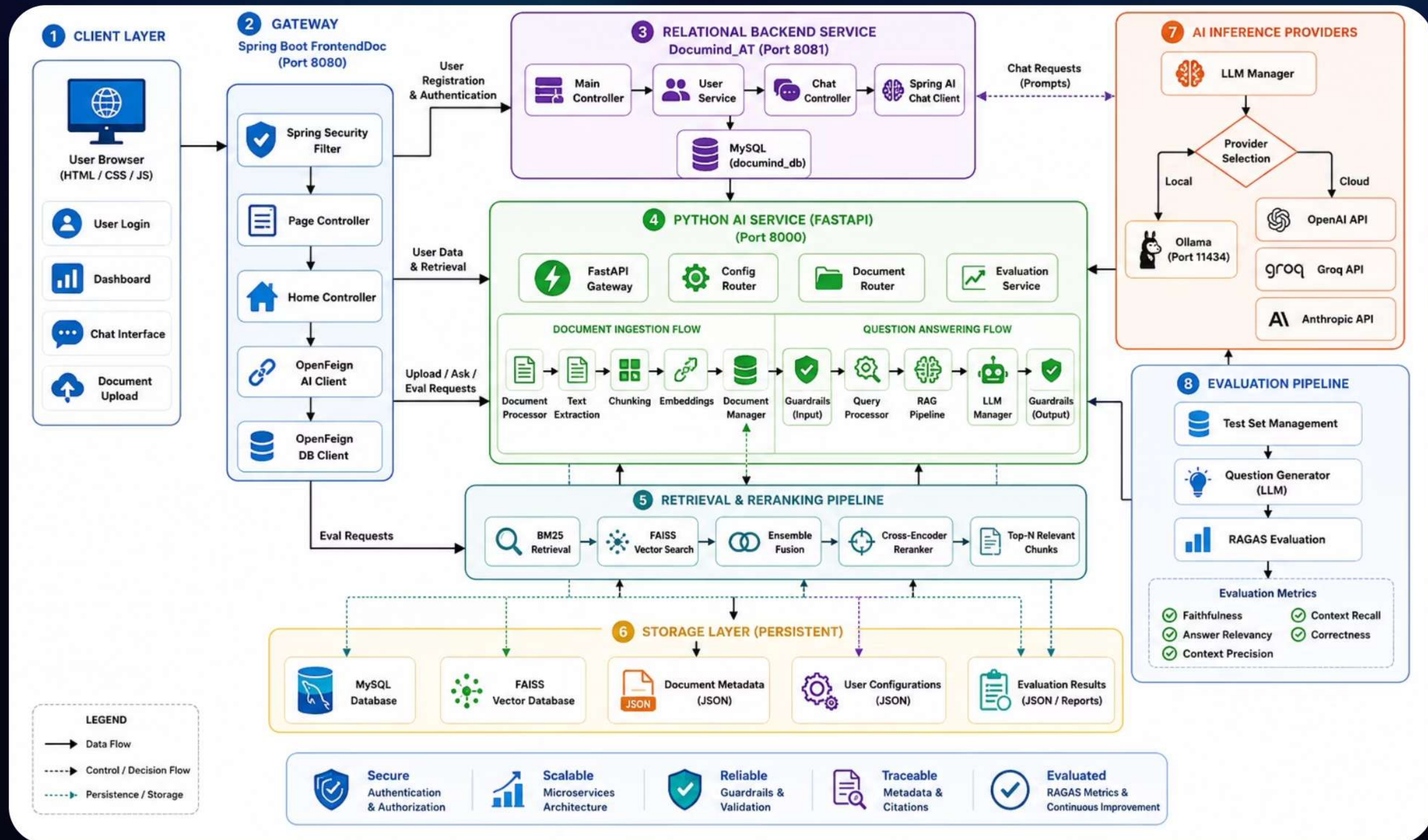


Multi-Tier Design

- **Client:** Responsive browser UI
- **Gateway:** Spring Boot routing & security
- **AI Layer:** FastAPI retrieval engine & LLM orchestration
- **Storage:** MySQL + FAISS vector store

End-to-End Data Flow

Hybrid RAG Knowledge Assistant – from user query to grounded, evaluated response



Document Ingestion Pipeline

Upload & Detect

PDF, DOCX, TXT,
CSV, Excel,
Markdown

Extract & Clean

Parse content,
normalize text

Store in FAISS

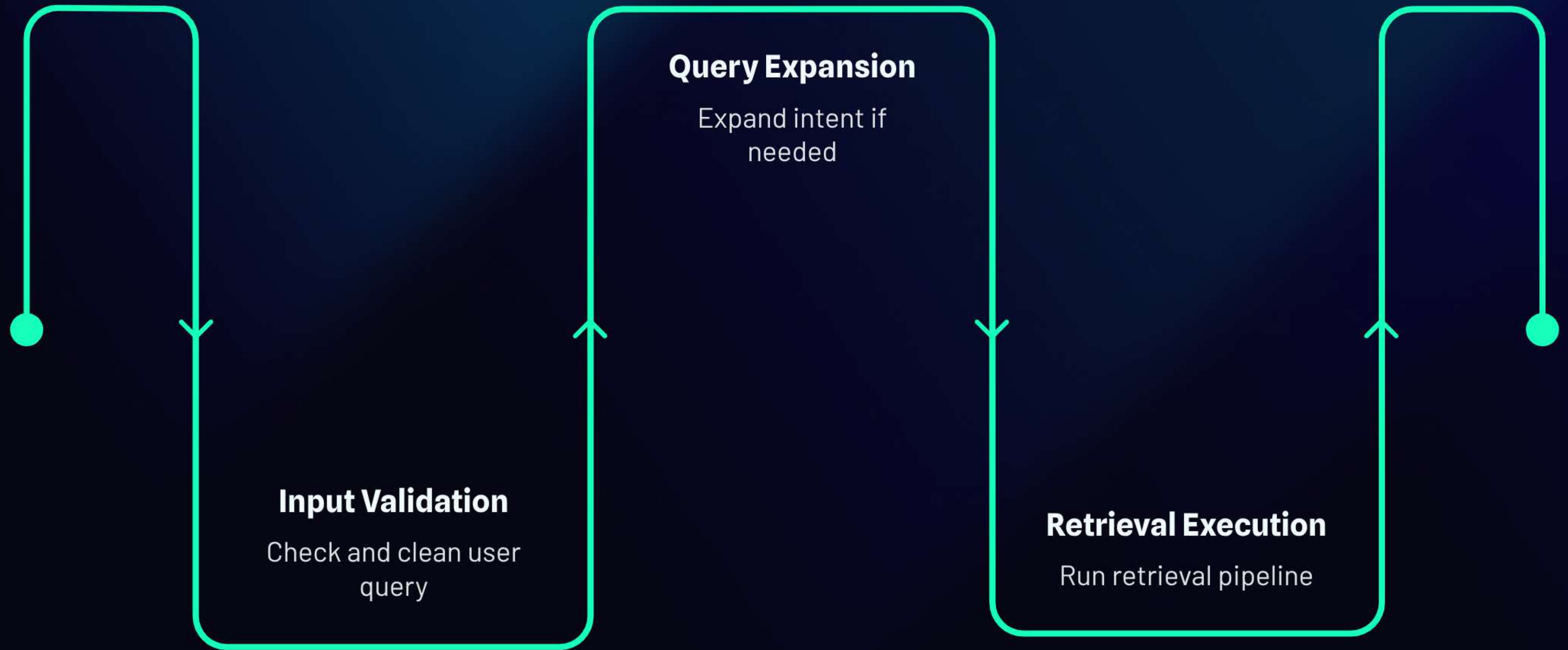
Index vectors for
retrieval

Chunk & Embed

Split into
segments,
generate
embeddings

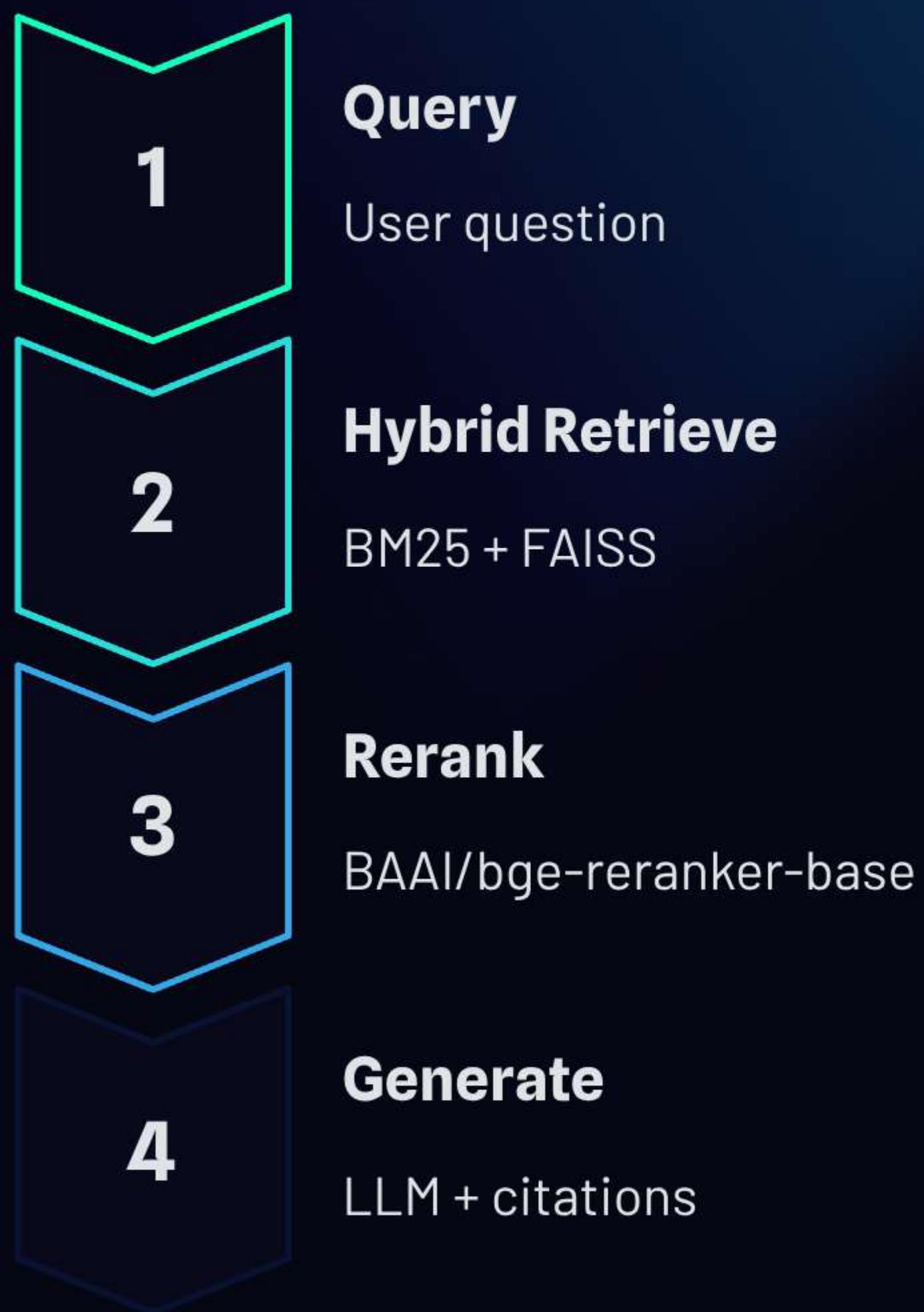


Adaptive Query Processing



Dynamic routing reduces latency and improves retrieval quality by classifying query complexity before execution.

Retrieval, Reranking & Response



Top chunks reranked by cross-encoder → LLM generates grounded, cited responses.

Results & Achievements



Reduced Hallucinations

Output guardrails and context checks significantly cut unsupported claims.



Source-Cited Responses

Every answer includes traceable references to original documents.



Secure & Private

Supports fully offline deployment with encrypted document management.

LOOKING AHEAD

Conclusion & Future Scope

Future Roadmap



Multimodal RAG

Images, audio, and
video retrieval



Agentic AI

Autonomous multi-
step workflows



Multi-Language

Global language
support



Enterprise Scale

Large-scale
organizational
deployment



Hybrid retrieval improves
answer quality



Guardrails ensure trustworthy
outputs



Scalable, privacy-first
architecture

Thank You

We welcome your questions, feedback, and discussion.

Project Team

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B.Tech CSE · Academic Session 2025–26